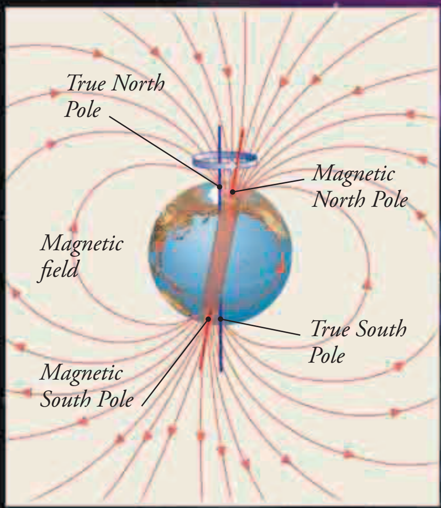
Magnetic Earth

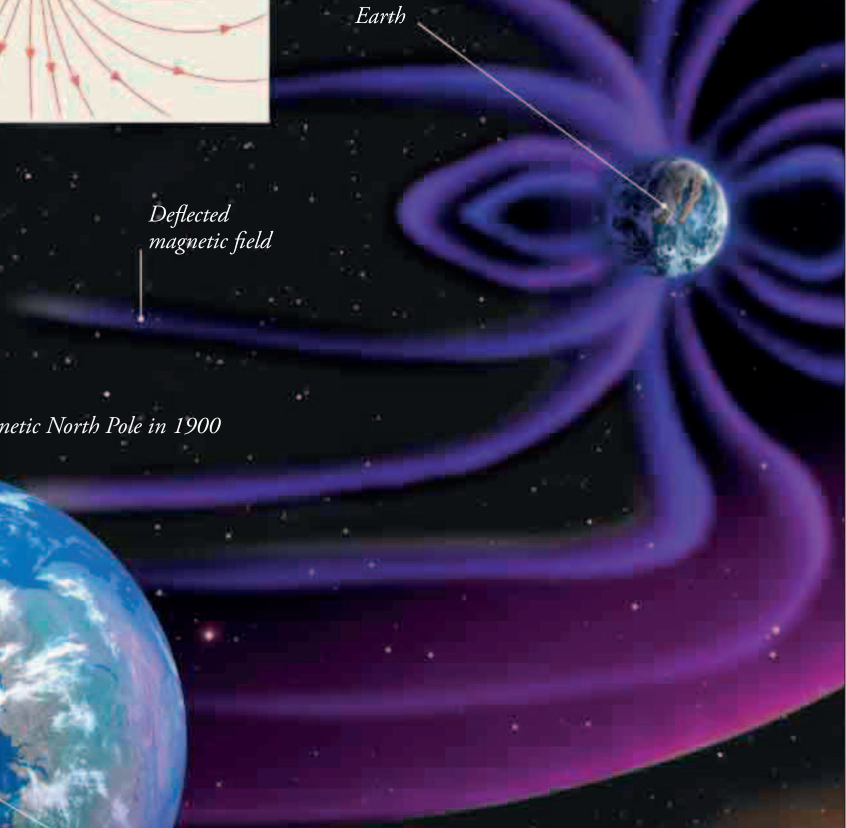
Earth's liquid metallic outer core is constantly in motion. The moving molten metal acts as an electromagnetic dynamo, generating a magnetic field around the planet. This is the same magnetism that makes a compass needle point north or south. More importantly, the magnetic field also deflects harmful rays from the Sun.

GEODYNAMO

The molten metallic outer core is kept in motion by heat currents and Earth's spin. This generates electrical currents in the iron-rich metal. In turn, these induce a magnetic field extending into space as the magnetosphere. It is similar to the field of a bar magnet roughly aligned with Earth's spin axis.



▸ OFFSET FIELD
The magnetic field is tilted at an angle to Earth's spin axis, so magnetic north is not true north.



WANDERING POLES

The strength of the magnetic field changes and so does its alignment with Earth's spin axis. This means that the magnetic North Pole keeps moving, at up to 25 miles (40 km) per year. Currently, it's very close to the true North Pole, but a century ago it was a long way south of it, in Arctic Canada. The magnetic South Pole also wanders, and the two magnetic poles are not always opposite each other.